Multiscale Short and Long Range Graph Convolutional Network for Hyperspectral Image Classification

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***Abstract*— Nowadays, graph convolutional networks (GCNs) are getting more attention in hyperspectral image classification (HSIC), and various algorithms based on GCNs have been proposed. However, because of hyperspectral images' (HSIs) complex spatial texture information, the long-range graph convolution (GConv) and short-range GConv may cause inaccurate or oversmoothed feature extraction of some nodes. Thus, a multiscale short- and long range graph convolutional network (MSLGCN) is proposed for HSIC. First, MSLGCN not only extracts spatial information of ground objects at different scales but also simultaneously captures global and local spectral features, which preserves objects' fine boundaries. Then, the rich multiscale information is complementary, enabling the MSLGCN to take full advantage of texture structures of varying sizes. In addition, a method to determine the superpixel scale by the intrinsic properties of HSIs is proposed to ensure that the segmentation boundary depicts the texture structure of the object accurately. Finally, the short–long graph convolution (SLGConv) is designed to fuse the advantages of global and local features, enabling the MSLGCN to extract accurate spatial–spectral features of nodes at any location. Experiments on three HSI datasets indicate that the MSLGCN can obtain better classification performance when compared with the other 11 state-of-the-art methods.**

***Index Terms*— Graph convolution (GConv), hyperspectral image classification (HSIC), multiscale, superpixel.**

I. Introduction

HYPERSPECTRAL images (HSIs) contain dozens or hundreds of spectral bands, which can provide more prosperous and distinguishable features than red, green, blue (RGB) images [1], [2]. Therefore, HSIs have been widely used in many fields [3], [4], including precision agriculture, military surveillance, and environmental monitoring. Moreover, hyperspectral image classification (HSIC) can provide pixel-level category information for subsequent tasks [5].

The machine learning (ML)-based algorithms use statistical learning and probabilistic methods to learn the intrinsic statistical patterns of an HSI and then construct the classification models. In detail, the common feature extractors can be principal component analysis (PCA) [6], linear discriminant analysis (LDA) [7], manifold learning [8], low-rank representation [9], etc. Note that the locality constraint criterion and structure preserving strategy were proposed to make full use of the low-rank representation in [9]. In addition, logistic regression (LR) [10], support vector machine (SVM) [11], and k-nearest neighbor (KNN) [12] can be considered as some of the classical classifiers.

Due to the powerful feature representation ability, the deep-learning-based methods have become the mainstream in the field of HSIC [13]. Specifically, the deep learning techniques can be categorized as: autoencoders (AEs) [14], deep belief networks (DBNs) [15], convolutional neural networks (CNNs) [16], deformable convolutional networks (DCNs) [17], [18], vision-transformer-based networks (ViTs) [19], and graph convolutional networks (GCNs) [20]. CNN can be seen as an effective method for extracting distinguishable features. For instance, Hu *et al.* [21] extracted spectral features of an HSI using a 1-D CNN (1-DCNN). Subsequently, 2-D CNN (2-DCNN) [22] and 3-D CNN (3-DCNN) [23] were introduced successively to extract spatial–spectral features, which mainly make use of the spatial smoothness of HSIs. Moreover, Roy *et al.* [24] combined 2-D and 3-D convolution to propose a hybrid spectral network called HybridSN.

However, the CNN-based algorithms have two main drawbacks. One is the inaccurate spatial information extraction due to the convolutional kernel's immutable shape. The other is that the local receptive field of the convolutional kernel cannot use the global information, which leads to inconsistency of the long-range intraclass features.

Generally, features refer to the discriminatory information that is used to distinguish categories, for example, spectral curve, spatial structure, texture, shape, and color. Furthermore, the features can be classified into long- and short-range features based on the distance between different pixels (features). Moreover, Fig. 1 illustrates the difference between local receptive field, long range, and

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short range. In brief, the long- and short-range node features can be understood as global and local features, respectively.

Graph learning [25] is widely used for various remote sensing tasks due to its good feature representation capability. For instance, Miao *et al.* [26] designed a random multigraphs' construction and ensemble strategy to achieve graph-based semisupervised HSIC. Afterward, Kipf and Welling [27] integrated the previous graph learning theory and proposed graph convolution (GConv). Nowadays, GCN is gaining more and more attention, which can compensate for the drawbacks of CNN for HSIC tasks. In contrast to CNN, GCN aggregates the features of neighboring nodes to achieve convolutional operations in the non-Euclidean space [28], which enables the GCN to encode the spectral information and spatial structure information of ground objects simultaneously [29]. The edge structure of the graph reflects the adjacency relationship between the global nodes, and GCN is naturally suitable for extracting long-range features [30]. Therefore, GCN is applied to encode HSI data with irregular texture features [31], [32]. Qin *et al.* [20] first proposed a GConv classification network to extract spatial–spectral information. Afterward, many GCN-based algorithms [33], [34] were proposed for HSI classification. Some of these algorithms considered each pixel of an HSI as a graph node [35], [36]. For instance, Mou *et al.* [36] proposed the nonlocal GCN, which took the whole HSI as input and achieved the extraction of long-range features. In addition, the deepening of the layers of the GCN will cause the deep features to be too smooth and lose the original features of the nodes, so Guo *et al.* [35] proposed a two-branch graph U-Net algorithm to relieve the oversmooth phenomenon.

The way that each pixel is considered as a node makes the graph highly complex, which can increase the computational complexity. The complexity reduction can be achieved by reducing the number of nodes. Thus, KNN [32], [37] or superpixel segmentation [25], [38], [39], [40] has been applied as a preprocessing step for clustering HSIs. In general, superpixel segmentation is considered a mainstream pixel-to-region approach. Furthermore, GCN has also been combined with other convolutional strategies [41], [42], [43], [44] to achieve good classification performance by incorporating the advantages of each. For example, a CNN enhanced GCN (CEGCN) hybrid network was proposed to extract the pixel-level and region-level features in [41], and another one is that the attention mechanism was introduced into a ten-layer deep GCN network to control edge weights in [42].

The above algorithms mainly carry out network-level researches for GConv, while some algorithms revolve around how to aggregate different ranges of node features. In brief, GConv can be divided into long- and short-range GConv, depending on the distance of node features, as shown in Fig. 1. The short-range GConv enables the aggregation of short-range node features within a few hops, as illustrated in Fig. 2(b) and (c). Note that the one-hop, two-hop, or three-hop GConv can be considered as short-range GConv. The long-range GConv can be understood that the features of all the nodes in the graph are simultaneously aggregated to

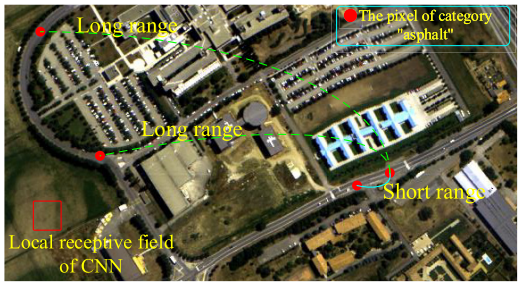


Fig. 1. Difference between local receptive field, long range, and short range. The red dots are the four pixel points of the category "asphalt," and the red square represents the local receptive field of CNN.

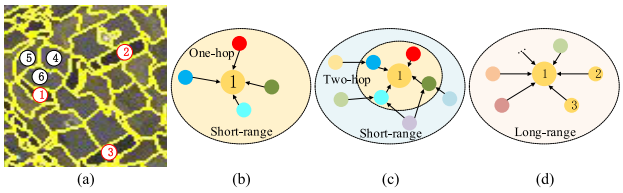


Fig. 2. (a) Position relationship of graph nodes. (b) Node aggregation for one-hop (short-range). (c) Node aggregation for two-hop (short-range). (d) Node aggregation for long-range.

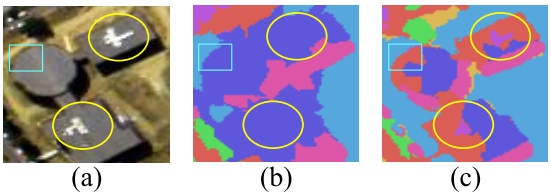


Fig. 3. Classification result maps of short- and long-range GConv. (a) Raw HSI. (b) Classification result map of short-range GConv. (c) Classification result map of long-range GConv.

the node ① to be aggregated in one aggregation operation, as shown in Fig. 2(a) and (d). For instance, the multiscale dynamic GCN (MDGCN) concatenated one-hop, two-hop, and three-hop neighbor node features to achieve multiscale feature fusion in [45], and the same approach can be seen in [18]. It can be concluded that the number of layers of GConv controls the range of node aggregation, i.e., the number of layers determines the number of hops [46]. In this line of thought, a multilayer GConv network is proposed to extract the long-range feature in [47].

In general, short-range GConv extracts local features, ensuring the accuracy of most node features, while long-range GConv extracts global features, which can improve the algorithm's robustness to cope with intraclass variability and interclass similarity of spectral curves [48]. Unfortunately, either long-range or short-range GConv has its drawbacks. To explain the limitations better, the classification result maps of long- and short-range GConv are shown in Fig. 3.

The short-range GConv cannot extract the global range features. Moreover, when the central node is surrounded by nodes of different categories, such as node ① in Fig. 2(a), the influence of the surrounding nodes on the node ① is significant, which will make node ① lose its feature after two or three layers of GConv, and finally leads to loss of boundary

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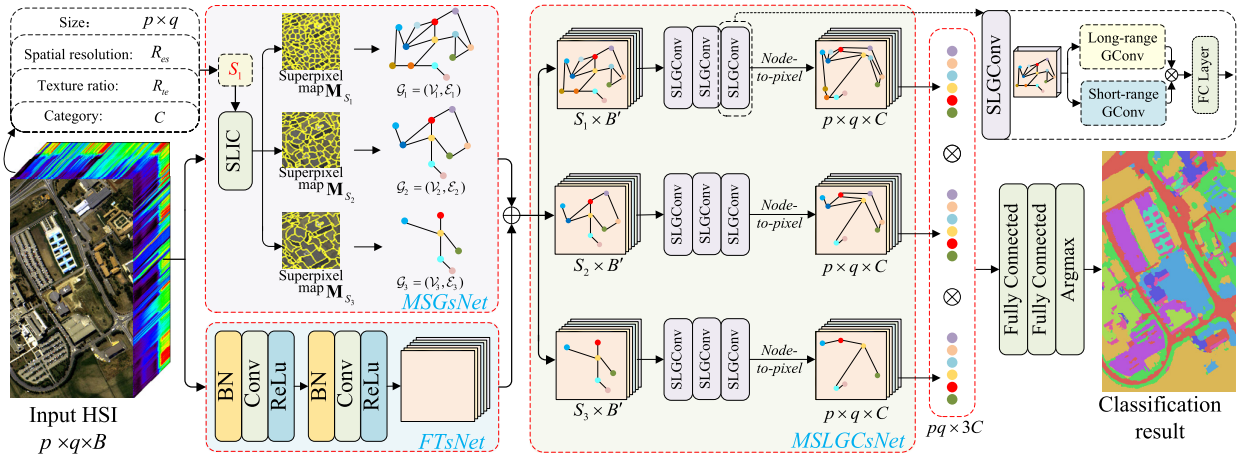


Fig. 4. Flowchart of the MSLGCN framework. ⊕ means combining the encoded graph nodes with their corresponding features. ⊗ denotes feature concatenation. "Node-to-pixel" indicates that the node-level features are converted into pixel level for classification, and the brief structure of SLGConv is shown in the upper right corner.

structure and oversmoothed classification results. Specifically, there are many oversmoothed regions in Fig. 3(b), especially the regions marked by circles. As for the long-range GConv, the features of node ① are the weighted average of global nodes, and although there will still be the influence of noncategory nodes, nodes ②③ of the same category have a greater weight than the noncategory nodes. Therefore, the original features of node ① will be preserved. However, when the node is surrounded by nodes of the same category, such as node ④ in Fig. 2(a), the features extracted by long-range GConv are not as accurate as those extracted by short-range GConv. Because long-range GConv inevitably aggregates impure information into node ④, while short-range GConv does not. Especially, the regions marked by rectangles in Fig. 3(c) indicate that the node features are not accurate.

In addition, due to the different sizes of ground objects in HSIs, a single-scale graph cannot capture the complex spatial structure [49] and may lose the boundary information of small objects. Furthermore, one graph may incompletely portray the boundaries of large objects, which eventually leads to the problem of oversmoothing or inaccurate feature extraction. The application of multiscale ideas [50], [51] can effectively handle tasks in complex scenes. For example, Liu *et al.* [52] achieved good performance in remote sensing object detection using multiscale features.

To cope with the above problems, a multiscale short and long range graph convolutional network (MSLGCN) is proposed for HSIC. The MSLGCN is designed with a new GConv approach that fully integrates the global and local features to ensure the purity of spatial–spectral features at each node and also uses three parallel branches to extract the spatial structure of objects at different scales. Meanwhile, a method to automatically determine the superpixel scale is also proposed to carve the objects' boundaries objectively. Note that the new GConv can make the boundaries of objects in the classification results consistent with the natural boundaries. The main contributions of this article are as follows.

1) First, a new MSLGCN is proposed. This framework uses three independent GConv processes to extract spatial–spectral features at different scales, ensuring that the texture structures of ground objects of different sizes are fully used. Meanwhile, each branch of GConv makes the MSLGCN take full advantage of labeled and unlabeled samples and preserve the precise boundary structures of the objects in the classification result maps.
2) Second, a new approach is proposed to determine the superpixel segmentation scale from the inherent properties of HSIs. This approach takes the HSI size, spatial resolution, texture ratio, and ground object category into account and can automatically calculate the superpixel scale instead of setting the scale manually.
3) Third, to extract global and local spatial–spectral features simultaneously, the short–long graph convolution (SLGConv) is designed, which uses parallel upper and lower branches to extract local and global features, respectively. The SLGConv combines the advantages of the two GConvs and eliminates the problems of oversmoothing of local features and inaccuracy of global features. Therefore, SLGConv can extract accurate global and local fused features and ensure classification performance.

The following structure of this article is as follows. Section II introduces the methodology of the proposed method. Section III presents the classification results and parameter discussion. Section IV summarizes this article.

II. Proposed Method

The flowchart of MSLGCN is shown in Fig. 4, which contains three subnetworks, multiscale superpixel generation subnetwork (MSGsNet), feature transformation subnetwork (FTsNet), and multiscale short–long-range graph convolution subnetwork (MSLGCsNet). First, the HSI is input into MSGsNet for multiscale superpixel segmentation, and the obtained superpixel regions at each of the three scales are encoded into graphs. Then, FTsNet is used to reduce the dimensionality of the raw HSI and eliminate noise, resulting in deep and robust spectral features. Subsequently, the three graphs and the features processed by FTsNet are combined to obtain node-level features, which are input into MSLGCsNet.

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TABLE I
DEFINITIONS OF NOTIONS USED IN MSGSNET AND FTSNET

Notation	Definition	Type	Size
$\mathcal{V}$	the node of the graph	concept	-
$\mathcal{E}$	the edge of the graph	concept	-
$\mathcal{G}$	the graph	concept	-
p, q, B	the length, width, band of HSI	scalar	1×1
C	the category of HSI	scalar	1×1
R_{es}	the spatial resolution of HSI	scalar	1×1
R_{te}	the texture ratio of HSI	scalar	1×1
S	the number of superpixels	scalar	1×1
S_n	the number of superpixels of nth scale	scalar	1×1
$\mathbf{m}_k(i)$	the ith pixel position of kth region	scalar	$(-,-)$
$\boldsymbol{x}_i$	spectral signature of ith pixel	vector	$1 \times B$
$\boldsymbol{V}_k$	the feature vector of the kth node	vector	$1 \times B'$
$\boldsymbol{W}^1_{\text{FTs}}$	the weight of first convolution of FTsNet	vector	$1 \times B$
$\mathbf{X}$	the HSI	matrix	$p \times q \times B$
$\mathbf{Y}$	the output result of MSLGCN	matrix	$p \times q$
$\mathbf{X}^{\text{out}}$	the output feature of FTsNet	matrix	$p \times q \times B'$
$\mathbf{M}_{S_n}$	superpixel segmentation map with S_n	matrix	$p \times q$
$\mathbf{m}_k$	the kth superpixel region	matrix	$p' \times q'$

Finally, MSLGCsNet extracts the spatial–spectral features at the three scales using the proposed SLGConv and obtains the final classification result. The detailed definitions of notions used in MSGsNet and FTsNet are shown in Table I.

A. Multiscale Superpixel Generation Subnetwork

The MSGsNet can be seen in Fig. 4, whose primary purpose is to generate undirected graphs $\mathcal{G}_i = (\mathcal{V}_i, \mathcal{E}_i), i = 1, 2, 3$ at three different scales (i.e., the number of superpixels), where $\mathcal{V}$ and $\mathcal{E}$ represent the node and edge of the graph, respectively. It is noted that LDA is used for supervised dimensionality reduction in the HSI before superpixel segmentation, which is inspired by [41].

Many studies [53], [54], [55], [56], [57] have shown that multiscale superpixel segmentation is considered as one of the effective strategies to cope with the underutilization of spatial information, which is caused by the complex structure, variable size, and interleaved distribution of ground objects in HSIs. However, in most of these algorithms, the determination of the superpixel scale is still based on human experience, which may make the segmentation map inconsistent with texture structures of ground objects.

Therefore, combining the previous research findings [58], a new method to determine the superpixel scale S from the inherent properties of HSI is proposed. This method takes the size $p \times q$, the spatial resolution R_{es}, the texture ratio R_{te}, and the category C of ground objects into consideration, which can be expressed as

$$S = \frac{p \times q \times R_{\text{te}} \times C}{100 \times 0.7^{\sqrt{R_{\text{es}}}}} \tag{1}$$

where R_{te} represents the ratio of the number of pixels N_{outline} at the object's outline to the total number of pixels N_{all} in the HSI, i.e., $R_{\text{te}} = N_{\text{outline}}/N_{\text{all}}$. Imitating [1], the Canny operator [59] is applied to calculate R_{te}.

In 1, it is easy to find out that the larger the size, the greater the number of object categories, and the higher the texture ratio, the greater the number of superpixel regions needed to segment the HSI. As for the spatial resolution, in the same scene, the change in spatial resolution does not change the texture scale and segmentation boundary of the image, and yet the size of the image becomes smaller as the spatial resolution increases, so that the superpixel scale S is proportional to $p \times q/0.7^{\sqrt{R_{\text{es}}}}$. As the scale S is calculated by 1, the number of superpixels at multiple scales can be set as

$$S_1 : S_2 : \ldots : S_n = S : S/k_2 : \ldots : S/k_n \tag{2}$$

where k_n is the scale factor used to control the number of superpixel regions at different scales, which is generally set to $k_n = n$, and n is the number of scales selected for use. Since the scale indicates the number of superpixel regions, we can treat $S_1 : S_2 : \ldots : S_n$ as different superpixel scales.

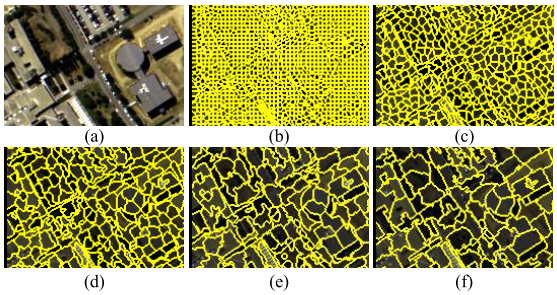


Fig. 5. Part of segmentation result maps of the PaviaU dataset. (a) Raw HSI. The number of superpixel scale is (b) 14091, (c) 4444, (d) 2258, (e) 1255, and (f) 869.

Note that the primary purpose of introducing multiscale is to fully use the spatial structure information of various ground objects in HSIs. Hence, the essential thing is that the superpixel segmentation maps at different scales should contain as much complementary structural information as possible. From the analysis of the results in Fig. 5, it seems that two scales cannot contain enough spatial structures, while five scales have many redundant structures. Therefore, it seems that choosing three and four scales is appropriate, and there are corresponding ablation experiments in Section III to further explain the influence of the number of scales.

At the end of MSGsNet, each superpixel region is considered as a node, and the positional relationship between superpixels (whether regions are adjacent or not) determines the node's edges (whether nodes are connected or not). Eventually, three graphs $\mathcal{G}_1, \mathcal{G}_2, and \mathcal{G}_3$ at different scales are obtained.

B. Feature Transformation Subnetwork

HSIs may be affected by the hardware or illumination during acquisition, so the spectral or spatial features may contain some undesirable noise. Moreover, the raw HSIs provide fine spectral features while containing much redundant information. To alleviate the problems, the FTsNet consisting of two layers of 1×1 convolution is designed. The FTsNet achieves the reduction in spectral dimension and extraction of deep spectral features, which further achieves the denoising of spectral features and the reduction in SLGConv computation. In the FTsNet, the process of convolution is expressed as

$$\begin{aligned} \mathbf{X}^1 &= \sigma\left(\boldsymbol{W}^1_{\text{FTs}} \cdot \text{BN}(\mathbf{X}) + \boldsymbol{b}^1\right) \\ \mathbf{X}^{\text{out}} &= \sigma\left(\boldsymbol{W}^2_{\text{FTs}} \cdot BN\left(\mathbf{X}^1\right) + \boldsymbol{b}^2\right) \end{aligned} \tag{3}$$

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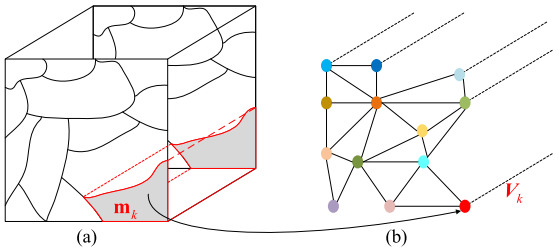


Fig. 6. Diagram of the conversion of pixel-level features to node-level features. (a) Pixel-level feature $\mathbf{X}^{\text{out}}$. (b) Node-level feature $\mathbf{X}^{S_1}$.

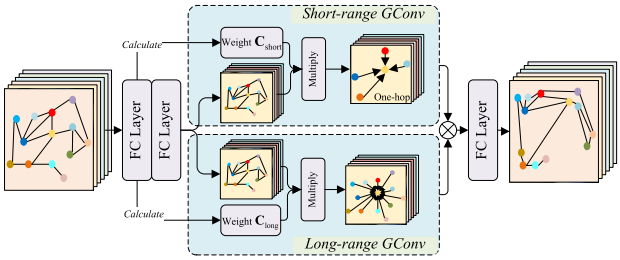


Fig. 7. Diagram of SLGConv. $\bigotimes$ denotes the feature concatenation, and $\mathbf{C}_{\text{long}}$ and $\mathbf{C}_{\text{short}}$ will be interpreted after 18.

where BN$(\bullet)$ and $\sigma(\bullet)$ denote the feature normalization and the activation function, respectively, $\mathbf{X}^1$ is the output of the first activation function, $\boldsymbol{W}^1_{\text{FTs}}$ and $\boldsymbol{b}^1$ denote the weight of the first convolution kernel and bias, respectively, and $\boldsymbol{W}^2_{\text{FTs}}$ and $\boldsymbol{b}^2$ denote the weight of the second convolution kernel and bias, respectively. The convolution does not change the size of the output feature, but only reduces the spectral dimension to B', which is expressed as $\mathbf{X} \in \mathbb{R}^{p\times q\times B} \rightarrow \mathbf{X}^{\text{out}} \in \mathbb{R}^{p\times q\times B'}$.

The output feature $\mathbf{X}^{\text{out}}$ is the pixel-level feature, which needs to be converted into the node-level feature for SLGConv. The core idea of the conversion is to average the features of all the pixels in each superpixel region; then the average feature is the node feature. The diagram of the conversion of pixel-level features to node-level features is shown in Fig. 6. In detail, assuming that a certain superpixel segmentation map $\mathbf{M}_{S_n}$ contains k superpixel regions $\{\mathbf{m}_1, \mathbf{m}_2, \ldots, \mathbf{m}_k\}$, and each region corresponds to a node in the graph, the process is formulated as

$$V_k = \frac{1}{n}\sum_{i=1}^{n} \mathbf{X}^{\text{out}}(\mathbf{m}_k(i)) \tag{4}$$

where $\mathbf{m}_k(i)$ denotes the ith pixel position of the kth superpixel region, $\mathbf{X}^{\text{out}}(\bullet)$ denotes the feature vector of the pixel at the corresponding position, and V_k represents the feature vector of the kth node. Finally, we can obtain node-level feature maps $\mathbf{X}^{S_1}, \mathbf{X}^{S_2}, \mathbf{X}^{S_3}$ at three different scales.

C. Short–Long-Range GConv

Although GConv has many unique advantages, it has not been applied on a massive level in the field of HSIC, probably because of the respective limitations of long- and short-range GConv in extracting features, as mentioned in Section I. There is no doubt that regardless of the form of GConv, the primary task is to ensure the correctness and purity of the extracted features, which can guarantee the accuracy of the classification results. From the previous analysis, it is known that short-range GConv is more suitable for feature extraction where all the nodes of the same category are in the local region, while long-range GConv is more suitable for extracting features of nodes located around nodes that are not of the same category.

The complex spatial structure of an HSI makes a considerable number of nodes adjacent to different classes of nodes. To extract the accurate features of all the nodes, we combine the advantages of long- and short-range GConv and propose SLGConv. The diagram of SLGConv is illustrated in Fig. 7.

TABLE II
DEFINITIONS OF NOTIONS USED IN MSLGCSNET

Notation	Definition	Type	Size
$\mathbf{H}$	the node matrix	matrix	$n \times B'$
$\mathbf{A}$	the adjacency matrix	matrix	$n \times n$
$\mathbf{D}$	the degree matrix	matrix	$n \times n$
$\mathbf{L}$	the graph Laplacian matrix	matrix	$n \times n$
$\mathbf{U}$	the eigenvector matrix of $\mathbf{L}$	matrix	$n \times n$
$\boldsymbol{u}$	the eigenvalue of $\mathbf{L}$	matrix	$n \times 1$
$\boldsymbol{\Lambda}$	the eigenvalue matrix of $\mathbf{L}$	matrix	$n \times n$
$\mathbf{g}_\theta$	filter parameter	matrix	$n \times n$
$\mathbf{S}$	the similarity matrix	matrix	$n \times n$
$\boldsymbol{f}, \boldsymbol{g}$	the given function	vector	$n \times 1$
$\boldsymbol{F}$	the GFT of $\boldsymbol{f}$	vector	$n \times 1$
$T_K(\bullet)$	K-order Chebyshev polynomials	-	-
$\mathcal{GF}[\bullet]$	the operation of GFT	-	-

The SLGConv consists of two branches at the top and bottom: short-range GConv and long-range GConv. Then the extracted features are concatenated together, and finally, a fully connected (FC) layer achieves weight control of the features of the top and bottom branches so that SLGConv can enable accurate feature extraction of different kinds of nodes. Note that the first FC layer can make the similarity relationship between each node more accurate, and the first two FC layers provide the convolution kernel parameters of SLGConv. In short, the core idea of SLGConv is that: by weighting the features extracted by fusing long-range GConv and short-range GConv, the advantages of both the convolutions are retained, and their respective disadvantages are compensated. The definitions of notions used in MSLGCsNet are shown in Table II.

The nodes $\mathcal{V}$ and edges $\mathcal{E}$ of the graph are represented in the matrix form, which are the node matrix $\mathbf{H} \in \mathbb{R}^{n\times B'}$ and adjacency matrix $\mathbf{A} \in \mathbb{R}^{n\times n}$, respectively, where n is the number of nodes, and B' is the dimension of node features. The degree matrix of the graph is $\mathbf{D} \in \mathbb{R}^{n\times n}$. Fig. 8 illustrates the physical meaning between these matrices. It can be seen that there are 12 superpixel regions in Fig. 8(a), corresponding to 12 nodes. The node matrix $\mathbf{H} \in \mathbb{R}^{12\times B'}$ stores the node features, and the adjacency matrix $\mathbf{A} \in \mathbb{R}^{12\times 12}$ stores the adjacent of nodes, e.g., nodes $\mathbf{m}_1$ and $\mathbf{m}_2$ are adjacent, then $\mathbf{A}(1, 2) = 1$, and the remaining positions are 0. The degree matrix $\mathbf{D} \in \mathbb{R}^{12\times 12}$ is a diagonal matrix, which stores the number of nodes adjacent to each other, $\mathbf{D}(i, i) = \sum_j \mathbf{A}(i, j)$.

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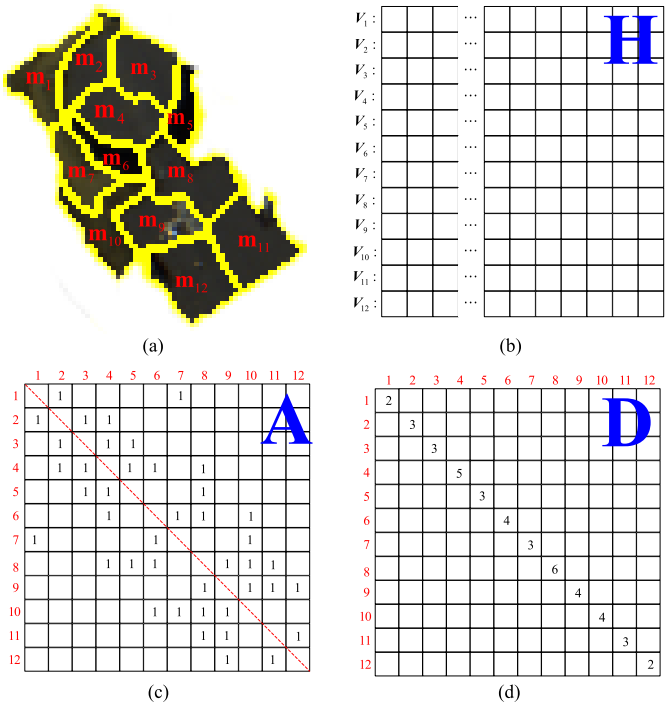


Fig. 8. Physical meaning between these matrices. (a) Superpixel segmentation result map. (b) Node matrix **H**. (c) Adjacency matrix **A**. (d) Degree matrix **D** of the graph.

From Fig. 8, it can be seen that the characteristics of the graph are represented in the degree matrix **D** and the adjacency matrix **A**. Then the graph Laplacian matrix [60] is defined as

$$\mathbf{L} = \mathbf{D} - \mathbf{A}. \tag{5}$$

Therefore, the characteristics of the graph are represented by the Laplace matrix, thus turning the graph's analysis into the Laplace matrix's analysis. Then, the Laplace matrix is normalized, i.e., $\hat{\mathbf{L}} = \mathbf{D}^{-1/2}\mathbf{L}\mathbf{D}^{-1/2}$. Meanwhile, for the convenience of presentation, the normalized Laplace matrix is still denoted as **L**. According to [61], **L** is a symmetric positive semidefinite matrix that can be spectral decomposition as

$$\mathbf{L} = \mathbf{I} - \mathbf{D}^{-1/2}\mathbf{A}\mathbf{D}^{-1/2} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^T \tag{6}$$

where $\mathbf{U} = [\boldsymbol{u}_1, \boldsymbol{u}_2, \ldots, \boldsymbol{u}_n] \in \mathbb{R}^{n\times n}$ and $\mathbf{\Lambda} \in \mathbb{R}^{n\times n}$ are the matrix consisting of the eigenvectors and eigenvalues of **L**, respectively. Note that the eigenvectors of **L** are orthogonal to each other, and $\mathbf{\Lambda}_{ii} = \lambda_i$, where λ_i is the eigenvalue. Thus, by regarding the eigenvectors of **L** as a set of basis $(\boldsymbol{u}_1, \boldsymbol{u}_2, \ldots, \boldsymbol{u}_n)$, the graph Fourier transform (GFT) of the given function $\boldsymbol{f} \in \mathbb{R}^{n\times 1}$ can be expressed as

$$\boldsymbol{F}(n) = \mathcal{GF}[\boldsymbol{f}] = \sum_i \boldsymbol{u}_n^T(i)\boldsymbol{f}(i) \tag{7}$$

where $\mathcal{GF}[\bullet]$ denotes the operation of GFT. Based on 7, the GFT can also be written in the form of the matrix: $\mathcal{GF}[\boldsymbol{f}] = \mathbf{U}^T\boldsymbol{f}$, and the inverse transform will be $\mathcal{GF}^{-1}[\boldsymbol{F}] = \mathbf{U}\boldsymbol{F}$, where $\mathcal{GF}^{-1}[\bullet]$ is the inverse GFT.

By referring to [30], the Fourier transform of the convolution of two functions $\boldsymbol{f}$ and $\boldsymbol{g}$ is the product of the Fourier transforms of $\boldsymbol{f}$ and $\boldsymbol{g}$, which can be formulated as

$$\mathcal{F}[\boldsymbol{f} * \boldsymbol{g}] = \mathcal{F}[\boldsymbol{f}] \cdot \mathcal{F}[\boldsymbol{g}]. \tag{8}$$

An alternative calculation of the convolution of $\boldsymbol{f}$ and $\boldsymbol{g}$ can be obtained by simultaneously applying an inverse Fourier transform $\mathcal{F}^{-1}[\bullet]$ to both sides of 8

$$\boldsymbol{f} * \boldsymbol{g} = \mathcal{F}^{-1}[\mathcal{F}[\boldsymbol{f}] \cdot \mathcal{F}[\boldsymbol{g}]]. \tag{9}$$

So imitating the convolution of the Fourier transform, the GConv of functions $\boldsymbol{f}$ and $\boldsymbol{g}$ is denoted as

$$\boldsymbol{f} *_{\mathcal{G}} \boldsymbol{g} = \mathbf{U}\big[\big(\mathbf{U}^T\boldsymbol{f}\big) \cdot \big(\mathbf{U}^T\boldsymbol{g}\big)\big] \tag{10}$$

where $*_{\mathcal{G}}$ represents the GConv, and $\cdot$ represents the point multiplication. If assume $\boldsymbol{g}$ is a filter [61], the filter parameter can be denoted as $\mathbf{g}_\theta = \mathrm{diag}(\mathbf{U}^T\boldsymbol{g})$, and 10 is represented as

$$\begin{aligned}\boldsymbol{f} *_{\mathcal{G}} \boldsymbol{g} &= \mathbf{U}\big[\big(\mathbf{U}^T\boldsymbol{f}\big) \cdot \big(\mathbf{U}^T\boldsymbol{g}\big)\big] = \mathbf{U}\big[\big(\mathbf{U}^T\boldsymbol{g}\big) \cdot \big(\mathbf{U}^T\boldsymbol{f}\big)\big] \\ &= \mathbf{U}\mathbf{g}_\theta\mathbf{U}^T\boldsymbol{f}.\end{aligned} \tag{11}$$

Then, GConv for node features can be expressed as

$$\boldsymbol{f}^{l+1} = \mathbf{U}\mathbf{g}_\theta\mathbf{U}^{\mathrm{T}}\boldsymbol{f}^l \tag{12}$$

where $\boldsymbol{f}^{l+1}$ and $\boldsymbol{f}^l$ represent the lth input and $l+1$th output features of GConv, respectively.

To reduce the computational cost, Hammond *et al.* [62] proposed a method of fitting $\mathbf{g}_\theta$ using the K-order Chebyshev polynomials $T_K(\bullet)$, and $\mathbf{g}_\theta$ is considered as the function with respect to $\mathbf{\Lambda}$, which is expressed as

$$\mathbf{g}_\theta(\mathbf{\Lambda}) \approx \sum_{k=0}^{K} \theta_k T_K(\tilde{\mathbf{\Lambda}}) \tag{13}$$

where $\tilde{\mathbf{\Lambda}} = 2\mathbf{\Lambda}/\lambda_{\max} - \mathbf{I}$, and furthermore, $\tilde{\mathbf{L}} = 2\mathbf{L}/\lambda_{\max} - \mathbf{I}$. In most cases, it is enough that K equals 1, so $\mathbf{g}_\theta(\mathbf{\Lambda}) = \theta_0 T_0(\tilde{\mathbf{\Lambda}}) + \theta_1 T_1(\tilde{\mathbf{\Lambda}})$. Since $T_0(\tilde{\mathbf{\Lambda}}) = 1, T_1(\tilde{\mathbf{\Lambda}}) = \tilde{\mathbf{\Lambda}}$, we get $\mathbf{g}_\theta(\mathbf{\Lambda}) = \theta_0 + \theta_1\tilde{\Lambda}$. If $\lambda_{\max} \approx 2$ and $\theta = \theta_0 = -\theta_1$, 12 can be finally represented as

$$\begin{aligned}\boldsymbol{f}^{l+1} = \mathbf{U}\big(\theta_0 + \theta_1\tilde{\mathbf{\Lambda}}\big)\mathbf{U}^{\mathrm{T}}\boldsymbol{f}^l &= \theta_0 h^l - \theta_1\mathbf{D}^{-1/2}\mathbf{A}\mathbf{D}^{-1/2}\boldsymbol{f}^l \\ &= \theta\big(\mathbf{I} + \mathbf{D}^{-1/2}\mathbf{A}\mathbf{D}^{-1/2}\big)\boldsymbol{f}^l.\end{aligned} \tag{14}$$

Notably, the similarity matrix **S** is computed using the attention mechanism idea to represent the similarity between each node. Meanwhile, it is known that the node features are stored in the matrix **H**, so we can get

$$\mathbf{S}(i,j) = \mathrm{Sig}\big(\big[\mathbf{W}_{\mathrm{fc}}^1\mathbf{H}(i)\big] \cdot \big[\mathbf{W}_{\mathrm{fc}}^1\mathbf{H}(j)\big]\big) \tag{15}$$

where $\mathbf{W}_{\mathrm{fc}}^1$ is the learnable parameters of the first FC layer in SLGConv, $\mathbf{H}(i)$ denotes the ith node feature, and $\mathrm{Sig}(x) = 1/(1+e^{-x})$, which normalizes x to between 0 and 1. In addition, the edges from each node to itself are added so that the node does not lose its own features after several layers of convolution operations, i.e., $\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}$, and correspondingly $\tilde{\mathbf{D}}(i,i) = \sum_j \tilde{\mathbf{A}}(i,j)$. Afterward, since **S** contains the similarity relationship between each node and all the remaining nodes, the adjacency matrix of the long-range GConv can be defined as

$$\tilde{\mathbf{A}}_{\mathrm{long}}(i,j) = \begin{cases}\mathbf{S}(i,j), & i \neq j \\ \mathbf{S}(i,j) + 1, & i = j.\end{cases} \tag{16}$$

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From Fig. 8(c), the adjacency between nodes is represented in $\mathbf{A}$. The short-range GConv only aggregates the features of adjacent nodes, so the adjacency matrix of the short-range GConv can be defined as

$$\tilde{\mathbf{A}}_{\text{short}}(i,j)=\begin{cases}\mathbf{S}(i,j)\cdot\tilde{\mathbf{A}}(i,j), & i\neq j\\ \mathbf{S}(i,j)\cdot\tilde{\mathbf{A}}(i,j)+1, & i=j.\end{cases}\tag{17}$$

Thus, based on the above analysis and 14–17, the SLGConv can be defined as

$$\mathbf{H}^{l+1}=\text{FC}\left[\sigma\left(\mathbf{C}^{l}_{\text{long}}\mathbf{H}^{l}\mathbf{W}^{12}_{\text{fc}}\right)\otimes\sigma\left(\mathbf{C}^{l}_{\text{short}}\mathbf{H}^{l}\mathbf{W}^{12}_{\text{fc}}\right)\right]\tag{18}$$

where FC[•] denotes the rightmost FC layer in SLGConv, $\sigma(\bullet)$ represents the activation function, $\otimes$ denotes feature concatenation, and $\mathbf{W}^{12}_{\text{fc}}$ are the parameters of the two FC layers on the left in Fig. 7. Note that the weights $\mathbf{C}_{\text{long}}$ and $\mathbf{C}_{\text{short}}$ can be represented as

$$\begin{aligned}\mathbf{C}_{\text{short}}&=\mathbf{I}+\tilde{\mathbf{D}}^{-1/2}\tilde{\mathbf{A}}_{\text{short}}\tilde{\mathbf{D}}^{-1/2}\\ \mathbf{C}_{\text{long}}&=\mathbf{I}+\tilde{\mathbf{D}}^{-1/2}\tilde{\mathbf{A}}_{\text{long}}\tilde{\mathbf{D}}^{-1/2}.\end{aligned}\tag{19}$$

D. Multiscale Short–Long-Range GConv Subnetwork

To fully exploit the spatial structure features, we propose MSLGCsNet using SLGConv and the idea of multiscale feature extraction. The MSLGCsNet consists of three parallel convolutional branches for extracting spatial–spectral features at different scales, and each branch is composed of three layers of SLGConv. The dimensions of the input node features of MSLGCsNet are $S_1 \times B'$, $S_2 \times B'$, and $S_3 \times B'$, where S_1, S_2 and S_3 are the number of nodes in the three scales, respectively, and B' is selected as 64 in this article. After the three-layer SLGConv, the node features are transformed into pixel-level features of dimension $p \times q \times B'$ by assigning the features of the nodes to each pixel point in that superpixel region, and this process is called "node-to-pixel."

The MSLGCN algorithm adopts the cross-entropy loss function to train the parameters, and we also apply category number weight, which is detailed in [31]. The loss function can be formulated as

$$\mathcal{L}=-\sum_{i=1}^{N}\sum_{c=1}^{C}N_c^{-1}\mathbf{O}_c(y_i)\log\left(\mathbf{Y}_c^{p}(\boldsymbol{x}_i)\right)\tag{20}$$

where N_c and N denote the sample number of category c and the number of total training samples, respectively. $\mathbf{O}_c(y_i)$ denotes the one-hot encoding format of y_i, and $\mathbf{Y}_c^p$ is the predicated probability of sample $\boldsymbol{x}_i$ that belongs to the cth category.

III. Experiments

To validate the performance of MSLGCN, three datasets are chosen to testify its classification accuracy, namely, the Pavia University (PaviaU) dataset [31], the Salinas dataset [1], and the Houston University 2013 (Houston2013) dataset [13]. Moreover, 11 state-of-the-art algorithms are chosen for comparison, namely, SVM [11], KNN [12], 2-DCNN, DCNet [17], double-branch dual-attention mechanism network (DBDA) [63], HybridSN [24], ViT [64], SpectralFormer [19], CEGCN [41], MiniGCN [30], and MDGCN [45]. The overall accuracy (OA), average accuracy (AA), and Kappa coefficient (Kappa) are used to quantitatively evaluate the performance of the algorithm, which are commonly known as evaluation indicators for HSI tasks [54].

TABLE III
Main Attributes of the Three HSI Datasets

Attributes	PaviaU	Salinas	Houston2013
Labels	42776	54129	4398
Size	610×340	512×217	344×552
Bands	103	204	144
Categories	9	16	10
Spatial resolution	$1.3m$	$3.7m$	$2.5m$
Scenario type	Urban	Farmland	Urban
Spectral range	0.43–0.86μm	0.4-2.5μm	0.38–1.05μm
Superpixel scales	4356 1581 1212	2316 793 574	4069 2029 1720

A. HSI Datasets

The basic information of these three datasets is shown in Table III, which lists the image size, spatial resolution, number of bands, and number of object categories. In particular, the Houston2013 dataset is a part (from rows 3 to 346, and from columns 635 to 1186) of the original Houston dataset, which contains most of the ground objects of the original dataset. Then, the number of superpixel regions of each dataset calculated by (1) is 4373, 2417, and 3871, respectively. It is important to note that these calculated numbers are only a guideline estimate, which are not the value to be strictly set. To facilitate the implementation, the number of superpixel regions for each dataset was set as shown in Table III.

B. Experimental Parameter Configuration

The MSLGCN is implemented in Python and the PyTorch framework with an initial learning rate of 0.0005, which uses the Adam optimizer. The number of training epochs of the MSLGCN algorithm is 600. The main hardware parameters for implementing all the algorithms are 1) central processing unit (CPU): Xeon Silver 4210R; 2) graphic processing unit (GPU): GTX-3090; and 3) RAM memory: 49G. The input patch size for the CNN-based algorithm is 17 × 17.

It is well-known that GCN, as a kind of semisupervised learning, can effectively use the information of unlabeled samples. So to verify the classification accuracy in the case of minimal training samples, 0.5%, 0.1%, and 99.4% of samples per class of the PaviaU and Salinas datasets, and 1%, 1%, and 98% of samples per class of the Houston2013 dataset are randomly selected as the training, validation, and testing datasets, respectively. Detailed category information of the three datasets is shown in Tables IV–VI. Furthermore, for a fair comparison, the number of training samples for each algorithm is standardized to be the same. The experimental data in this section are obtained by repeating the experiment ten times in parallel.

C. Analysis of Classification Results

This section compares and analyzes the classification performance of all the algorithms. The three superpixel segmentation

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TABLE IV
CATEGORY INFORMATION OF THE PAVIAU DATASET

Class ID	Class Name	Train.	Val.	Test.
1	Asphalt	34	7	6590
2	Meadows	94	19	18536
3	Gravel	11	3	2085
4	Trees/Grass	16	4	3044
5	Metalsheets	7	2	1336
6	Baresoil	26	6	4997
7	Bitumen	7	2	1321
8	Bricks	19	4	3659
9	Shadows	5	1	941
Total		219	48	42509

TABLE V
CATEGORY INFORMATION OF THE SALINAS DATASET

Class ID	Class Name	Train.	Val.	Test.
1	Brocoli_green_weeds_1	11	3	1995
2	Brocoli_green_weeds_2	19	4	3703
3	Fallow	10	2	1964
4	Fallow_rough_plow	7	2	1385
5	Fallow_smooth	14	3	2661
6	Stubble	20	4	3935
7	Celery	18	4	3557
8	Grapes_untrained	57	12	11202
9	Soil_vinyard_develop	32	7	6164
10	Corn_senesced_green_weeds	17	4	3257
11	Lettuce_romaine_4wk	6	2	1060
12	Lettuce_romaine_5wk	10	2	1915
13	Lettuce_romaine_6wk	5	1	910
14	Lettuce_romaine_7wk	6	2	1062
15	Vinyard_untrained	37	8	7223
16	Vinyard_vertical_trellis	10	2	1795
Total		279	62	53788

TABLE VI
CATEGORY INFORMATION OF THE HOUSTON2013 DATASET

Class ID	Class Name	Train.	Val.	Test.
1	Healthy grass	7	7	657
2	Stressed grass	3	3	233
3	Road	4	4	343
4	Trees	7	7	621
5	Soil	5	5	482
6	Water	4	4	331
7	Residential	3	3	229
8	Commercial	5	5	469
9	Parking Lot 1	6	6	544
10	Parking Lot 2	5	5	391
Total		49	49	4300

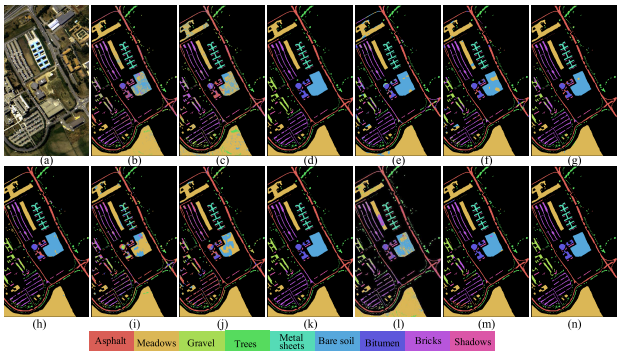


Fig. 9. Classification maps of all different methods on the PaviaU dataset. (a) False-color image. (b) Ground truth. (c) SVM. (d) KNN. (e) 2DCNN. (f) DCNet. (g) DBDA. (h) HybridSN. (i) ViT. (j) SpectralFormer. (k) CEGCN. (l) miniGCN. (m) MDGCN. (n) MSLGCN.

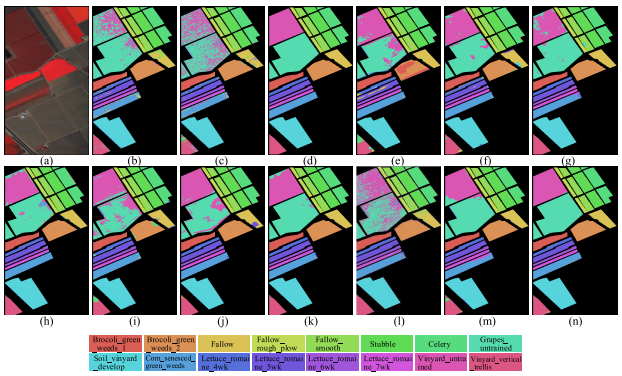


Fig. 10. Classification maps of all different methods on the Salinas dataset. (a) False-color image. (b) Ground truth. (c) SVM. (d) KNN. (e) 2-DCNN. (f) DCNet. (g) DBDA. (h) HybridSN. (i) ViT. (j) SpectralFormer. (k) CEGCN. (l) miniGCN. (m) MDGCN. (n) MSLGCN.

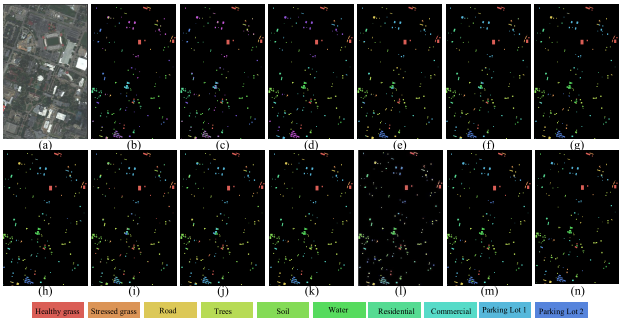


Fig. 11. Classification maps of all different methods on the Houston2013 dataset. (a) False-color image. (b) Ground truth. (c) SVM. (d) KNN. (e) 2-DCNN. (f) DCNet. (g) DBDA. (h) HybridSN. (i) ViT. (j) SpectralFormer. (k) CEGCN. (l) miniGCN. (m) MDGCN. (n) MSLGCN.

scales selected by MSLGCN are shown in bold in Table III. Figs. 9–11 show the classification result maps of the algorithms on the three datasets, respectively. Correspondingly, the classification accuracy of each category is also presented in detail in Tables VII–IX, respectively.

For the PaviaU dataset, the MSLGCN achieves the best overall classification accuracy with 0.5% training samples, which reaches 98.78%. As seen in Table VII, the classification accuracy of the CNN-based algorithms is more diminutive than 98% due to the small number of training samples, and only DBDA with the attention mechanism achieves the highest OA of 97.65%. Similarly, the transformer-based algorithm only achieved less than 80% of the classification accuracy. On the contrary, the GCN-based algorithms with semisupervised training have obtained better classification accuracy. Among them, CEGCN achieves the OA of 98.53% due to its strategy of combining CNN and GCN. Most importantly, the MSLGCN achieves the highest classification accuracy using only the GConv approach, which validates the effectiveness of the proposed SLGConv and multiscale feature extraction.

To be specific, it can be seen from Fig. 9(n) that the classification result map has hardly any region-level misclassification or pepper–salt noise. Because SLGConv extracts

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TABLE VII
CLASSIFICATION ACCURACY (%) OF ALL THE ALGORITHMS ON THE PAVIAU DATASET

Class	ML-based		CNN-based				Transformer-based		GNN-based			
	SVM	KNN	2DCNN	DCNet	DBDA	HybridSN	ViT	SpectralFormer	CEGCN	MiniGCN	MDGCN	MSLGCN
1	85.04±5.25	76.73±0.53	94.36±2.65	92.7±3.17	95.89±2.5	95.67±2.21	70.65±8.77	85.47±6.7	**98.84±1.26**	79.95±7.65	98.1±1.08	98.45±1.47
2	93.86±3.07	92.06±4.88	98.23±1.39	97.03±1.69	99.35±0.38	99.82±0.15	89.66±7.6	93.06±5.58	**99.91±0.22**	90.86±3.11	99.99±0	99.59±0.41
3	52.51±15.22	28.44±16.45	59.57±6.45	68.08±6.91	96.26±3.61	88.03±5.95	38.1±7.18	38.09±9.1	88.31±10.33	70.99±7.45	77.03±5.8	**99.06±1.46**
4	75.09±6.50	49.68±16.63	91.61±3.78	86.18±4.68	98.44±1.02	82.7±3.25	73.99±2.79	79.45±6.37	**93.84±2.15**	89.77±0.83	90.63±0.88	93.48±1.99
5	99.24±0.31	98.34±0.62	97.82±2.44	97.93±3.47	99.13±1.02	99.9±0.19	75.63±8.3	82.32±6.89	**100±0**	99.33±0.28	98.85±0.42	99.73±0.33
6	65.59±11.38	30.88±15.06	89.89±13.59	94.2±3.83	99.35±0.6	98.19±1.13	51.95±17.19	54.87±14.08	99.44±1.69	69.99±7.55	97.47±0.67	**99.84±0.3**
7	40.01±18.39	65.02±7.35	53.95±18.73	78.57±7.64	99.29±1.37	93.03±7.75	44.69±16.26	57.67±12.73	**99.72±0.51**	67.82±2.35	97.79±3.84	98.86±1.98
8	79.1±4.66	83.13±5.60	89.68±8.45	96.38±1.99	89.82±4.21	80.59±5.3	67.5±0.4	56.16±9.44	98.41±1.83	65.6±17.17	92.1±2.7	**99.43±0.52**
9	99.68±0.13	99.41±0.59	84.46±9	79.3±10.59	98.81±1.26	80.22±16.24	21±16.66	50.37±5.95	98.27±1.99	**99.89±0.09**	97.24±0	92.11±6.23
OA	83.15±1.30	75.08±1.19	91.84±2.54	92.83±1.14	97.65±0.56	94.88±0.75	73.36±2.26	78.16±1.99	98.52±0.79	83.24±2.61	96.76±0.57	**98.78±0.37**
AA	76.68±3.59	69.30±4.18	84.4±4.03	87.85±1.29	97.37±0.45	90.9±2.24	59.24±4.29	66.38±1.24	97.42±1.41	81.58±1.5	94.35±1.11	**97.84±0.91**
Kappa	77.31±1.89	65.81±2.31	89.09±3.55	90.52±1.49	96.88±0.75	93.16±1.01	64.17±2.35	70.37±2.35	98.03±1.05	77.91±3.16	95.69±0.75	**98.38±0.49**

TABLE VIII
CLASSIFICATION ACCURACY (%) OF ALL THE ALGORITHMS ON THE SALINAS DATASET

Class	ML-based		CNN-based				Transformer-based		GNN-based			
	SVM	KNN	2DCNN	DCNet	DBDA	HybridSN	ViT	SpectralFormer	CEGCN	MiniGCN	MDGCN	MSLGCN
1	96.79±1.30	93.97±0.98	82.81±24.62	94.66±4.76	99.71±0.55	**100±0**	90.95±7.69	97.13±3.09	99.91±0.26	98.9±0.19	99.5±0	99.88±0.36
2	97.65±1.69	98.01±0.28	84.5±19.85	92.39±6.53	99.97±0.05	**100±0**	93.19±0.76	98.2±1.93	**100±0**	99.29±0.24	99.92±0.19	**100±0**
3	87.19±7.97	85.46±9.28	87.78±8.99	84.07±4.5	96.32±0.24	99.82±0.17	85.48±9.51	89.81±7.53	**100±0**	59.49±18.87	100±0	99.48±1.56
4	95.43±3.56	98.73±0.36	97.13±4.15	96.51±2.28	90.89±2.65	89.42±7.14	95.32±2.23	94.91±3.36	99.28±0.69	98.72±0.74	**99.65±0.13**	98.97±2.05
5	95.83±2.41	96.66±0.82	94.93±4.22	97.56±1.74	99.33±0.3	**99.77±0.41**	98.12±1.3	98.99±1.43	99.19±0.64	96.47±2.82	97.88±0.02	97.97±0.7
6	98.80±0.37	98.96±0.25	99.43±0.98	99.52±0.6	99.03±0.69	99.7±0.45	99.78±0.06	98.25±1.46	**99.95±0.14**	99.78±0.02	99.55±0.12	99.79±0.44
7	99.25±0.32	98.73±0.43	96.42±6.82	96.77±1.97	99±1.6	99.9±0.06	97.32±2.71	98.96±0.71	**99.99±0.02**	99.48±0.07	95.61±3.74	99.88±0.13
8	88.48±1.18	74.41±5.24	87.47±18.15	86.13±8.22	96.53±3.49	93.09±1.22	68.54±5.38	82.1±1.02	97.4±2.53	70.59±5.08	95.39±2.11	**99.63±0.42**
9	96.95±0.83	96.91±1.52	97.22±2.68	98.3±1.18	99.39±0.19	99.95±0.09	98.12±2.51	99.97±0.04	**100±0**	99.64±0.24	99.74±0.35	**100±0.00**
10	83.39±4.13	76.64±5.51	95.64±2.13	95.56±2.66	96.28±2.15	98.99±0.68	95.8±1.91	95.52±0.42	96.21±0.88	92.13±1.22	89.51±2.52	**96.38±1.69**
11	79.65±4.42	72.93±17.44	87.83±13.92	85.79±13.13	89.8±3.71	93.04±3.15	86.92±8.4	94.51±3.33	**99.68±0.48**	94.66±1.49	95.76±1.13	97.98±2.08
12	99.04±1.40	98.33±1.17	82.31±24.81	94.09±6.79	99.45±0.66	**99.98±0.04**	97.2±2.55	99.1±1.28	99.98±0.05	99.67±0.36	100±0	99.81±0.38
13	97.47±1.06	96.76±1.96	94.77±5.34	93.97±2.29	99.34±0.55	96.24±2.41	94.52±2.22	97.95±0.98	**99.51±0.65**	78.38±4.55	98.79±0	97.32±3.78
14	88.05±6.86	90.35±2.13	95.76±3.99	95.7±2.68	89.78±3.89	94.72±3.43	97.22±2.52	94.18±8.23	**98.14±1.53**	83.36±5.31	94.66±1.41	97.89±1.5
15	53.086±5.47	52.21±5.07	48.6±34.21	88.8±6.9	84.11±11.6	91.42±1.7	91.51±1.88	88.86±6.24	98.3±1.25	66.92±5.31	91.47±0.61	**98.83±0.88**
16	88.47±5.68	72.58±11.05	89.31±6.09	91.55±2.46	**99.9±0.2**	99.63±0.41	75.19±13.01	99±1.38	99.52±0.43	97.15±1.35	95.19±2.98	97.2±6.21
OA	87.64±0.61	83.48±0.81	85.75±5.01	92.42±1.6	95.11±1.86	96.7±0.43	88.77±2.21	93.2±0.82	98.87±0.64	86.18±1.36	96.4±0.43	**99.15±0.37**
AA	90.35±0.35	87.60±2.014	88.87±4.24	93.21±1.5	96.18±0.7	97.23±0.79	91.57±2.32	95.46±0.55	**99.19±0.3**	89.66±1.96	97.04±0.34	98.81±0.75
Kappa	86.18±0.69	81.57±0.93	84.09±5.56	91.57±1.77	94.57±2.06	96.34±0.47	87.54±2.44	92.43±0.92	98.74±0.72	84.65±1.51	95.99±0.47	**99.05±0.42**

both the global and local spatial–spectral features to ensure that the surrounding information of different categories does not smooth out the information of each superpixel region (node). Meanwhile, the strategy of multiscale feature extraction applied by MSLGCN also ensures correct classification of ground objects of different sizes. On the contrary, the classification result maps of the rest of the comparison algorithms show more or less region-level misclassification or pepper–salt noise. In general, MSLGCN extracts long- and short-range spatial–spectral features at different scales to ensure the accuracy of the extracted node features, which in turn achieves high accuracy in the classification of each superpixel region.

The MSLGCN obtained the highest classification accuracy and the optimal classification result map for the Salinas dataset. Overall, the OA of MSLGCN reaches 99.15%, and the classification result maps of each category can reproduce the actual distribution of ground objects. In addition, analysis of the classification results of all the algorithms shows that the classification results of the category "Grapes untrained" are most prone to misclassification of small regions, most of which are confused with the category "Vinyard untrained." However, the MSLGCN corrects this problem, mainly due to the graph nodes at different scales containing different spatial information. In the subsequent fusion, if the features at one of the three scales fuse the wrong information, the remaining scales can still provide the correct and discriminative category information for the final classification. Table VIII shows that the classification accuracy of CEGCN is very close to that of MSLGCN; however, the MSLGCN achieves higher classification accuracy by just using GConv, which powerfully demonstrates the ability of SLGConv.

For the Houston2013 datasets, MSLGCN performed the optimal classification performance compared with the remaining algorithms. As seen in Fig. 11(a), the Houston2013 dataset is a challenging classification scenario with a staggered distribution and ground objects of varying sizes. However, MSLGCN achieves a classification accuracy of 88.42% with 49 training samples from ten categories, which demonstrates the ability of the MSLGCN to classify complex HSI. Meanwhile, each algorithm in Table IX shows a significant decrease in accuracy. In comparison, the classification performance of the GCN-based algorithm does not degrade as much as that of the CNN-based or transformer-based algorithms, mainly because the GCN-based algorithm uses unlabeled data. In addition, MSLGCN fuses global and local features to mitigate the effect of insufficient training samples.

D. Discussion of Experimental Parameters

In this section, first, the validity of the SLGConv is verified. Then, the influence of different numbers of training samples on the classification performance of each algorithm is discussed.

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TABLE IX
CLASSIFICATION ACCURACY (%) OF ALL THE ALGORITHMS ON THE HOUSTON2013 DATASET

Class	ML-based		CNN-based				Transformer-based		GNN-based			
	SVM	KNN	2DCNN	DCNet	DBDA	HybridSN	ViT	SpectralFormer	CEGCN	MiniGCN	MDGCN	MSLGCN
1	98.09±0.86	96.77±2.00	97.57±2.06	89.42±3.97	97.53±4.34	96.3±4.01	91.78±4.13	94.49±6.12	**98.93±1.1**	98.29±0.07	96.46±1.78	91.92±6.35
2	85.30±7.68	81.18±4.25	40.86±24.87	45.19±12.11	81.9±22.89	60.42±8.12	48.94±18.24	37.41±14.36	68.54±22.59	89.12±6.62	**91.32±6.66**	86.61±11.03
3	29.54±26.00	54.08±13.13	29.03±13.13	67.09±13.34	69.42±16.37	37.13±5.46	36.59±12.64	47.7±13.04	86.45±8.5	76.27±9.97	78.89±12.65	**89.71±5.52**
4	97.08±2.13	96.15±3.36	96.9±2.63	84.06±5.31	**97.91±1.56**	90.01±7.76	75.63±16.5	67.99±12.81	97.54±3.08	94.32±1.25	96.14±2.45	95.62±3.7
5	97.74±4.14	88.78±8.76	**99.32±1.36**	86.89±8.21	90.51±8.15	97.21±2.24	78.21±18.43	93.92±6.42	93.78±10.07	99.18±0.77	97.38±1.63	99.13±2.34
6	84.77±6.11	82.08±8.54	70.63±7.38	56.12±13.77	**94.48±7.28**	73.33±11.15	25.99±7.38	37.3±3.69	86.22±7.5	85.07±0.24	82.31±6.61	88.07±8.17
7	74.96±23.56	75.62±17.67	38.7±16.7	63.09±10.53	72.07±18.74	65.07±12.28	20.31±3.25	28.47±6.89	70.78±23.88	84.91±20.43	89.01±14.1	94.06±16.69
8	38.54±20.45	24.81±8.96	49.12±10.9	39.88±8.29	64.55±33.53	**72.28±5.43**	27.3±12.66	27.16±12.98	51.94±14.67	50.07±7.46	64.08±5.44	52.05±7.51
9	69.81±13.66	64.76±7.64	83.91±10.41	86.38±11.88	79.71±14.02	82.11±0.73	82.58±14.37	89.53±2.2	86.7±11.18	27.94±11.53	73.64±12.03	**90.94±9.88**
10	12.55±7.81	16.47±7.39	57.81±11.63	85.91±7.88	76.02±8.81	37.13±3.09	25.69±25.45	21.32±6.3	85.28±5.98	34.35±5.56	91.73±1.09	**94.91±3.33**
OA	71.61±4.13	70.07±2.59	73.25±5.05	74.1±1.67	82.85±8.31	75.76±0.87	58.09±4.7	61.2±0.58	85.12±3.44	73.86±2.38	86.5±0.17	**88.42±2.77**
AA	68.84±5.15	68.07±2.89	66.39±6.51	70.4±1.18	82.3±8.54	71.22±1.77	51.3±5.69	54.53±0.75	82.62±4.2	73.95±2.83	86.1±0.93	**88.3±3.16**
Kappa	67.96±4.71	66.22±2.98	69.63±5.82	70.81±1.85	80.74±9.26	72.59±1.02	52.27±5.58	55.98±0.65	83.22±3.89	70.71±2.67	84.83±0.17	**86.99±3.1**

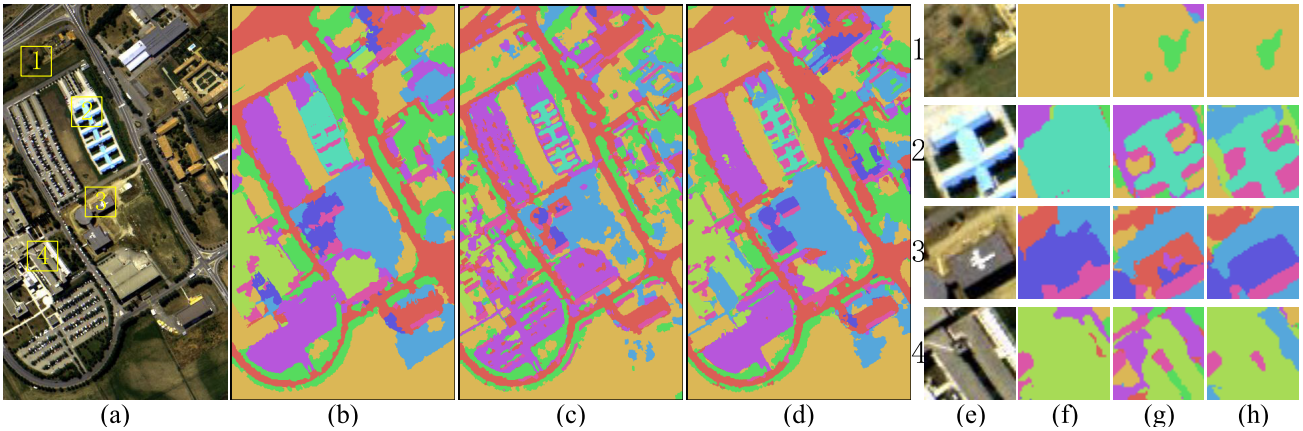


Fig. 12. Four enlarged regions in the classification result map on the PaviaU dataset. (a) False-color image. (b) Short-range GConv. (c) Long-range GConv. (d) SLGConv. (e) Enlarged region of (a). (f) Enlarged region of (b). (g) Enlarged region of (c). (h) Enlarged region of (d).

TABLE X
OA% OF EACH BRANCH OF SLGCONV

Dataset	Upper branch (short-range)	Lower branch (long-range)	SLGConv (short-long range)
PaviaU	97.49±0.78	95.75±0.79	**98.78±0.37**
Salinas	99.06±0.28	96.1±1.0	**99.15±0.37**
Houston2013	85.36±2.82	83.09±3.62	**88.42±2.77**

Moreover, the influence of the number of scales and the depth of MSLGCN on the classification performance is analyzed. Finally, the training time and network parameters (NPs) are compared for each algorithm.

1) Influence of the SLGConv: SLGConv is proposed as a new form of GConv, and the roles of its upper and lower branches are verified and analyzed. As introduced in Section II-C, the upper branch of SLGConv is the short-range GConv, and the lower branch is the long-range GConv. In the feature fusion process, the upper and lower branches are responsible for the feature extraction tasks at long and short ranges, respectively. Table X presents the OA of each part of SLGConv. The validation experiments used the completed framework of MSLGCN, except for replacing SLGConv with long- and short-range GConv. In addition, the NPs are set consistent with the parameters used in Section III-A.

As shown in Table X, SLGConv achieved much higher classification accuracy than either branch. For the Houston2013 dataset, the OA of SLGConv is improved by 3.06% and 5.33% compared with that of short- and long-range GConv, respectively. Moreover, the results in Table X show that SLGConv overcomes the drawbacks of the above two GConvs. Specifically, the way that two parallel branches extract features independently allows SLGConv to cope with nodes in different locations flexibly. When the central node is mainly surrounded by nodes of the same category, the weight of the features extracted by the upper branch (short-range GConv) increases; when the central node is mainly surrounded by nodes of different categories, the weight of the features extracted by the lower branch (long-range GConv) increases. Therefore, the features extracted by SLGConv have the characteristics of accuracy and purity, which can improve the classification accuracy. To illustrate the advantages of SLGConv in more detail, Figs. 12–14 shows the classification result maps of the MSLGCN framework using the short-range GConv, long-range GConv, and SLGConv, respectively.

These classification result maps clearly verify the drawbacks of long- and short-range GConv, as expressed in Section I. It can be summarized as follows: one is that short-range GConv is more likely to cause feature oversmoothing because it fuses features in a local region, which makes isolated or small-sized nodes be smoothed out. The other is that

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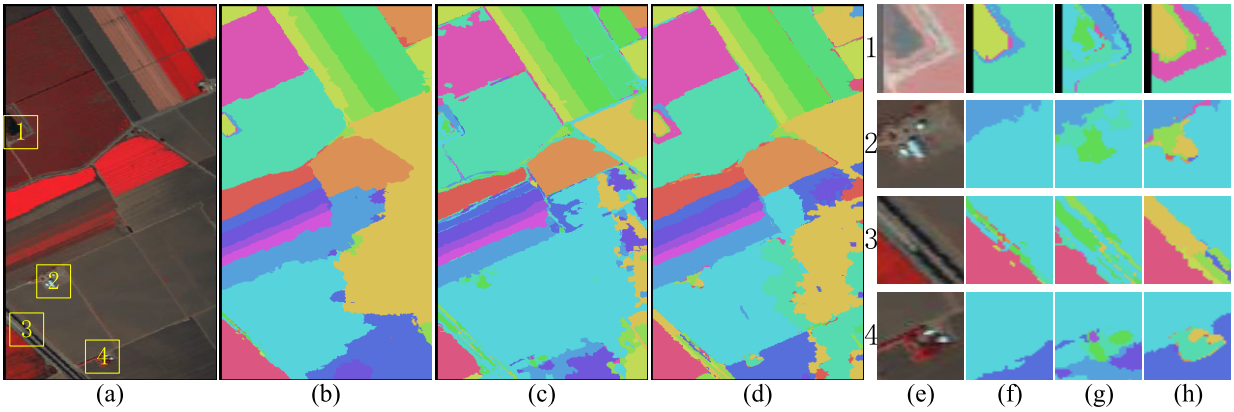


Fig. 13. Four enlarged regions in the classification result map on the Salinas dataset. (a) False-color image. (b) Short-range GConv. (c) Long-range GConv. (d) SLGConv. (e) Enlarged region of (a). (f) Enlarged region of (b). (g) Enlarged region of (c). (h) Enlarged region of (d).

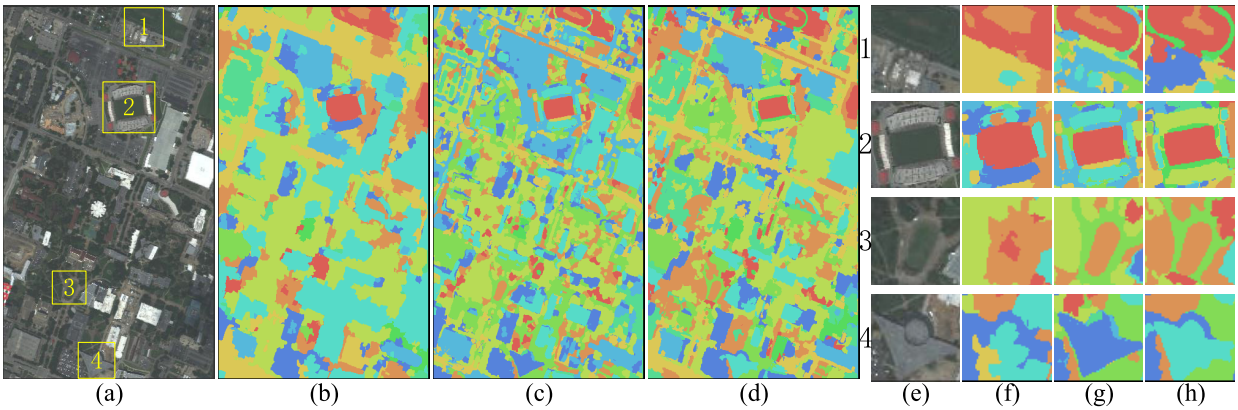


Fig. 14. Four enlarged regions in the classification result map on the Houston2013 dataset. (a) False-color image. (b) Short-range GConv. (c) Long-range GConv. (d) SLGConv. (e) Enlarged region of (a). (f) Enlarged region of (b). (g) Enlarged region of (c). (h) Enlarged region of (d).

long-range GConv inevitably brings in node features of different categories when features are extracted globally, resulting in inaccurate node features, so the classification results may have more misclassification phenomena. In summary, the enlarged region maps [Figs. 12–14(f)] show that the classification results of the short-range GConv lose most of the subtle boundary structures of the ground objects; Figs. 12–14(g) reveal that the shapes of the objects in the classification results of the long-range GConv remain consistent with the natural shapes of the objects, while the classification results are not very accurate. Correspondingly, the classification results of SLGConv not only restore the texture information of the object but also ensure the accuracy of the classification, as shown in Figs. 12–14(h), which verifies the superiority of SLGConv.

2) Influence of the Training Sample Size: Currently, the number of training samples has a significant impact on the classification performance of the algorithm. To verify the classification accuracy of MSLGCN under very few training samples, 2, 5, 10, 15, and 20 samples from each category are selected as the training dataset, respectively. Furthermore, one from each category is chosen as the validation dataset. The average overall accuracies of each algorithm are shown in Fig. 15.

It can be seen that compared with the remaining nine algorithms, MSLGCN attained the highest classification accuracy on the three datasets. Moreover, the fewer the training samples, the more pronounced the superiority of MSLGCN performance. In particular, under the condition that there are only two training samples per category, the OA of MSLGCN still reaches 73.59% on the PaviaU dataset. The data in Fig. 15 indicate that MSLGCN is very promising in handling HSIC tasks with insufficient label samples, mainly benefiting from the ability of SLGConv to fuse global and local features, i.e., the label information of a particular local node is easily propagated to the global level, so that the features of distant nodes also receive the corresponding labels. Since CEGCN combines the pixel-level classification result, it also achieves a classification accuracy close to that of MSLGCN.

3) Influence of the Number of Scales: The default number of scales adopted by the MSLGCN framework is three. However, it is conceivable that the number of scales adopted by the algorithm is also a crucial parameter, so the ablation experiments are designed to examine the classification performance of MSLGCN at one, two, three, and four scales. Point out that the number of training samples is shown in Section III-A, and the four superpixel scales for each dataset derived from 1 and 2 are 4356-2206-1581-1212 (PaviaU), 2316-1155-793-574 (Salinas), and 4069-3119-2029-1720 (Houston), respectively. More importantly, the four scales are numbered as $S_1, S_2, S_3, and S_4$ according to the size of

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TABLE XI

RUNNING TIME (S) AND NUMBER OF PARAMETERS OF ALL THE ALGORITHMS

Dataset	Times	CNN-based				Transformer-based		GNN-based			
		2DCNN	DCNet	DBDA	HybridSN	ViT	SpectralFormer	CEGCN	MiniGCN	MDGCN	MSLGCN
PaviaU	Training	43	45.61	74.67	40.57	859.51	883.11	42.86	95.89	7.51	56.06
	Testing	4.23	4.44	13.94	23.72	5.01	5.07	4.58	19.39	1.31	0.02
	NP	123.79K	879.75K	202.77K	4017.40K	174.45K	175.21K	152.86K	15.19K	6.81K	1547.45K
Salinas	Training	112	44.57	166.56	50.77	1090.73	1090.17	24.03	173.54	3.57	30.01
	Testing	7.17	5.43	33.73	29.76	6.01	6.35	2.75	42.48	0.10	0.01
	NP	153.78K	881.15K	389.64K	4018.30K	174.90K	175.66K	166.89K	29.22K	13.31K	892.94K
Houston2013	Training	24	38.78	27.85	22.2	94.72	102.72	40.74	5.08	3.25	57.31
	Testing	0.99	0.52	1.92	2.82	0.53	0.52	4.19	0.15	0.06	0.01
	NP	135.72K	879.95K	279.48K	4017.53K	174.51K	175.27K	158.32K	20.65K	9.33K	1662.35K

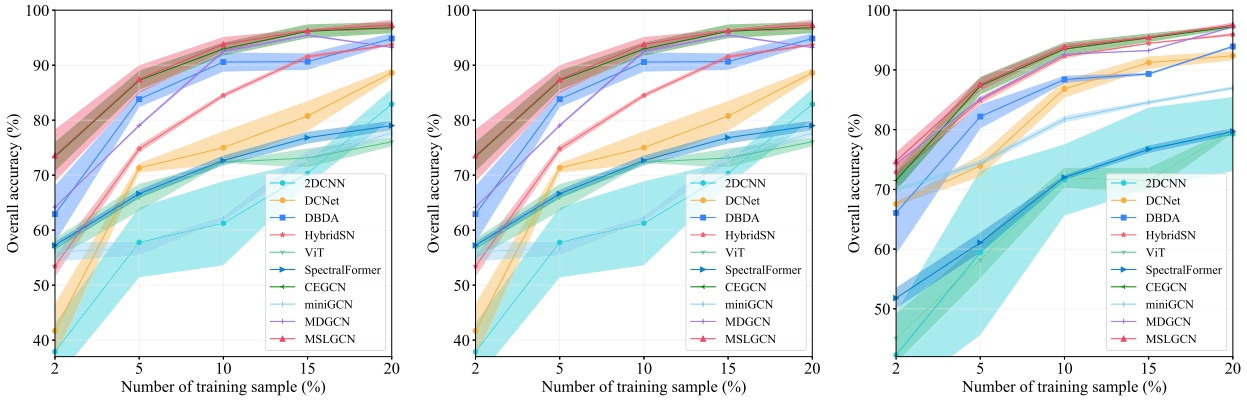


Fig. 15. Mean and standard deviation of the OA of all the algorithms under different numbers of training samples, where the width of the shaded part represents the size of the standard deviation and the solid line represents the mean value of OA. (a) PaviaU. (b) Salinas. (c) Houston2013.

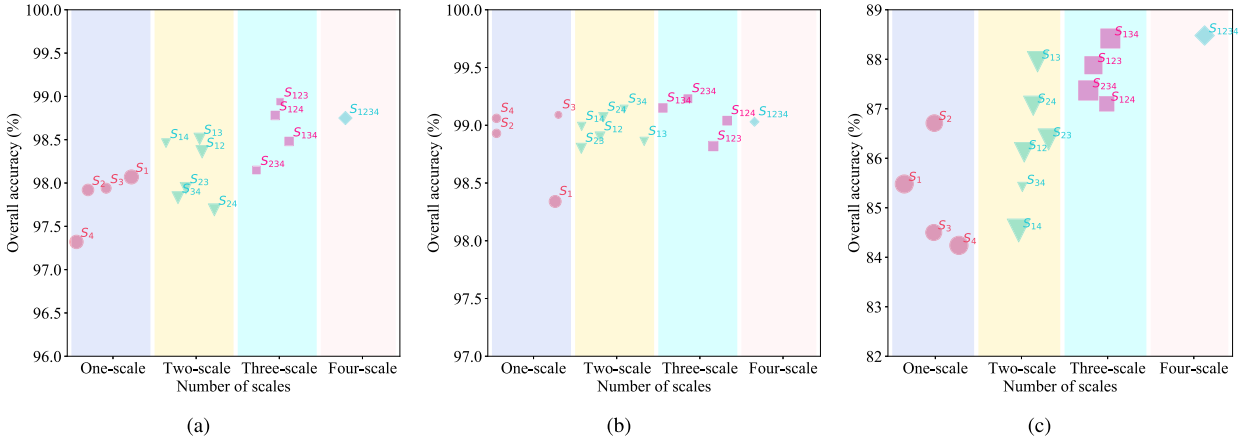
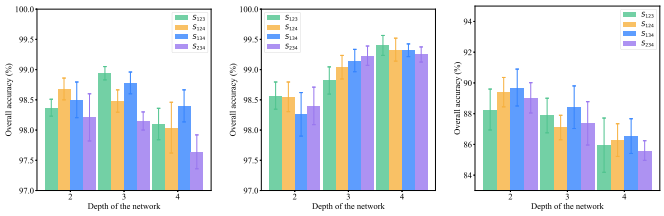


Fig. 16. Classification accuracy of the MSLGCN algorithm for 15 combinations of four superpixel scales, where the position of the vertical axis corresponding to the center of the geometric shapes is the classification accuracy and the area of each shape indicates the covariance of the OA. (a) PaviaU. (b) Salinas. (c) Houston2013.

the scale, from the largest to the smallest. Fig. 16 shows the classification accuracy of the MSLGCN algorithm for 15 combinations of four superpixel scales. Specifically, S_{123} denotes that $S_1, S_2, and S_3$ are selected as the three superpixel scales of MSLGCN. Note that when the number of scales is four, a new feature extraction branch needs to be added to MSLGCN. For a particular dataset, the data in Fig. 16 show a trend that the OA of MSLGCN becomes higher as the number of scales increases, mainly because the introduction of spatial information at multiple different scales makes the final node features more accurate, so the accuracy will improve. However, in a certain case [a certain color in Fig. 16(a), (b), or (c)], the combination of different superpixel scales also changes the classification accuracy, which can be understood

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(a) (b) (c)

Fig. 17. Classification accuracy of the MSLGCN algorithm for multiple scale combinations at different network depths, where the line segment on each bar indicates the covariance, and the longer it is, the larger the covariance is. (a) PaviaU. (b) Salinas. (c) Houston2013.

as the spatial structure information contained at different scales may be complementary, inclusive, or conflicting with each other. Therefore, a higher OA will be obtained if a certain combination of superpixel scales is in a complementary relationship. Overall, when the number of scales is 3, it can obtain satisfactory accuracy on all the three datasets, so the default number of scales of MSLGCN is set to 3. Meanwhile, the highest classification accuracy appears in different combinations of scales because the three datasets have different essential properties. This experiment illustrates the influence of the number of scales on the classification performance and also shows which combination of scales is optimal.

4) Influence of the Depth of the Network: In this section, the influence of the depth of the network is analyzed. The OA of multiple combinations of scales at different network depths is depicted in the histogram, as shown in Fig. 17. In this experiment, the number of superpixel scales is fixed to three, and the number of training samples is set according to Section III-A. It is well-known that the number of convolutional layers in a graph convolutional network should not be too many; otherwise, it will cause the problem of oversmoothing of features. So we verified the classification performance when using two to four layers of SLGConv.

In general, MSLGCN achieves high accuracy for network depths of 2, 3, and 4, respectively. Specifically, since the distribution of ground objects in the Houston2013 dataset is more complex, the deepening of the network depth increases the probability of nodes losing their own information and makes the node features smoother, so the maximum value of OA appears when the network depth is two layers. Similarly, on the PaviaU and Salinas datasets, the maximum value of OA occurs when the network depth is two and three layers, respectively, depending on the complexity of the datasets. The data in Fig. 17 show that MSLGCN does not show a sharp decline in OA as a result of the deepening of the network layers, which strongly demonstrates the ability of MSLGCN to handle datasets with different characteristics. Alternatively, SLGConv can cope with the problem of feature inaccuracy caused by the deepening of the network, because the spatial–spectral features which fuse global and local information are more able to reveal the categories of nodes.

5) Analysis of the FTsNet: In this section, the role of FTsNet is analyzed, and the OA of FTsNet with different parameter settings is shown in Table XII. As mentioned in Section II-B, FTsNet can reduce the original HSI's dimensionality and reduce the computational burden, while also enabling denoising. It can be seen that with the addition of FTsNet, the classification accuracy of MSLGCN improves significantly, for example, by 1.71% on the PaviaU dataset. On the other hand, the classification accuracy of FTsNet without reducing the HSI dimension ($B' = B$) is slightly lower than that of dimensionality reduction ($B' = 64$), indicating that dimensionality reduction can extract features and reduce noise. Meanwhile, smaller dimensionality ($B' = 32$) will cause the node features to lose some of the original information, which is demonstrated by a decrease in classification accuracy.

TABLE XII
OA% OF FTSNET WITH DIFFERENT PARAMETER SETTINGS

Dataset	Without FTsNet $B' = B$	With FTsNet $B' = B$	With FTsNet $B' = 64$	With FTsNet $B' = 32$
PaviaU	96.81±0.91	98.53±0.43	**98.78±0.37**	98.06±0.72
Salinas	98.26±0.73	98.88±0.53	**99.15±0.37**	98.60±0.57
Houston2013	85.48±2.13	86.57±3.35	**88.42±2.77**	86.12±3.72

6) Analysis of Runtime and NPs: In this section, the performance parameters of each algorithm are compared, e.g., training time, testing time, and the number of NPs, as shown in Table XI. As can be seen, MSLGCN has a comparable training time, thanks to the training approach of GConv, and also makes the testing time of the MSLGCN algorithm the least. The NPs of MSLGCN are more numerous, one is because SLGConv is a combination of two GConvs, and the other is because MSLGCN is composed of three branches, each of which in turn is composed of three layers of SLGConv. However, larger number of network training parameters does not drag down the algorithm's running time; on the contrary, MSLGCN achieves higher classification performance with comparable running time compared with the rest of the algorithms. The GCN-based algorithms have a faster training and testing time, while MSLGCN not only retains this advantage of GCN but also obtains the optimal classification performance.

IV. CONCLUSION

In this article, an MSLGCN is proposed for hyperspectral imagery classification. First, the MSLGCN can extract spatial information at different scales and also capture both the global and local spectral features of ground objects, which can enhance the classification accuracy and performance of the network. Second, the feature extraction portion of MSLGCN is composed of three parallel branches, which enables MSLGCN to make full use of the rich spatial information at different scales and thus obtain correct classification results for objects of different sizes. Furthermore, the superpixel scale is determined by considering the intrinsic properties of an HSI, which can reduce the impact of the inaccurate scale given by human experience on the classification results. Finally, as far as we know, the SLGConv is first proposed to extract simultaneously long- and short-range spatial–spectral features. In this way, SLGConv achieves feature fusion under global

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and local receptive fields and can highlight either of these two features to ensure the accuracy of the extracted features of nodes at different locations, which solves the problem of oversmoothing or inaccurate feature extraction that occurs in GConv. The experimental results on three datasets show that MSLGCN can achieve the best classification performance compared with other 11 algorithms.

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